Customer Stories / Software & Internet

2024

Achieving over 99.99% Resiliency Using AWS Services with Sprinklr

Learn how Sprinklr enhanced its customer experience solution by rearchitecting on AWS, improving resiliency using AWS.

Overview | **Opportunity** | **Solution** | **Outcome** | **AWS Services Used**

99.99%

resiliency

30%

cost reduction for key value workloads using Amazon Keyspaces

80%+

reduction in p90 and p99 latencies

Overview

To expand its offerings, Sprinklr created Sprinklr Service—a digital-first, proactive customer experience contact-center-as-a-service (CCaaS) solution. To meet Sprinklr Service customers' increased resiliency needs, Sprinklr rearchitected internally and used a stack of solutions from Amazon Web Services (AWS) to achieve high availability, data integrity, and optimal performance.

Opportunity | Using AWS Managed Services to Enhance CCaaS Solution Resiliency for Sprinklr

Founded in 2009, Sprinklr provides a customer experience solution to help businesses gain insights and track marketing through social media. The company has expanded over the years to help engage customers across more than 30 digital channels. It now offers four robust product suites that provide various customer experience and marketing tools, such as marketing analytics and social listening. In 2022, the company launched the Sprinklr Service product suite. When building this CCaaS solution, Sprinklr quickly realized that its customer base required increased resiliency.

To take full advantage of Sprinklr Service, customers needed high availability—and expected it. Although Sprinklr already had a resiliency of 99.9 percent, which worked well before these new offerings were implemented, that did not meet its new service-level agreement obligations of more than 99.99 percent resiliency. The company was born on AWS but needed to change its architecture, both on AWS and internally, to meet these new expectations.

One of the internal changes was the implementation of horizontal and vertical auto scaling to handle changing demands. In addition to that, Sprinklr implemented key resiliency patterns, like circuit breaker, timeout, bulkhead, and rate limiters, across all microservices to handle faults gracefully and prevent system overloads. Sprinklr also reduced the blast radius of backend services in terms of exposure to other components to minimize the scope of impact. One major point of rearchitecting that Sprinklr made on AWS was to migrate key value use cases to Amazon Keyspaces (for Apache Cassandra)—which is a scalable, highly available, and managed Apache Cassandra–compatible database service—to achieve scalability and resiliency. Sprinklr also uses Amazon ElastiCache—which offers near real-time performance for near real-time applications—to enhance performance.

Over 2 years, Sprinklr enhanced its solution using a combination of architectural best practice [illegible] WS services. This step-by-step rollout facilitated a 99.99 percent uptime for critical workloads and services with virtually no interruptions for customers. “We were confident that if we moved to managed AWS services and made these architectural changes, we’d achieve the resiliency we needed,” says Abhay Bansal, senior vice president of engineering at Sprinklr.

Solution | Reaching over 99.99 Percent Resiliency and Reducing Costs Using AWS

Sprinklr uses a variety of managed AWS services, including Amazon Elastic Kubernetes Service (Amazon EKS)—which is the most trusted way to start, run, and scale Kubernetes—to manage its Kubernetes workloads. “Migrating to Amazon EKS was pivotal for us,” says Nitin Goyal, vice president of engineering at Sprinklr. “We could scale up to abnormal workloads in essentially no time.” Sprinklr also uses Amazon Relational Database Service (Amazon RDS), which is an easy-to-manage relational database service optimized for total cost of ownership, and secures and stores its data on Amazon Simple Storage Service (Amazon S3), which provides object storage built to retrieve virtually any amount of data from anywhere.

The backbone of Sprinklr’s current architecture on AWS is Amazon Elastic Compute Cloud (Amazon EC2), which offers secure and resizable compute capacity for virtually any workload. Sprinklr has always been an early adopter of new Amazon EC2 instances, which has empowered it to make use of innovations and scalable compute capacity. For example, the company runs some of its Amazon ElastiCache workloads and 80 percent of its Amazon EKS and Amazon Keyspaces workloads on Amazon EC2 instances powered by latest AWS Graviton processors, which offer the best price performance for cloud workloads running on Amazon EC2, to enhance compute performance.

Sprinklr also uses Amazon EC2 I4i instances, which are powered by third-generation Intel Xeon Scalable processors, to boost its input/output performance. Using I4i instances, Sprinklr can run sustained peak MongoDB workloads with lower latencies and one-tenth the occurrences of downtime.

By changing backend infrastructure and using Amazon Keyspaces, Sprinklr reduced infrastructure degradations from three or four events per quarter to almost zero. “We reduced costs by 30 percent and still had 99.99 percent resiliency,” says Bansal. “Using Amazon Keyspaces, we achieved almost zero degradation with no operational overhead.”

Sprinklr has not only exceeded customer resiliency requirements but also reduced latency using AWS. Customer-observed tail latencies have decreased by over 80 percent. With the rearchitected infrastructu
customers have access to fast, highly available, and secure services.

Outcome | Embracing New Technologies to Support Customers

Sprinklr is always looking to make its offerings as smooth as possible for its customers, and by using new AWS services and new Amazon EC2 instances, its products are becoming even more resilient and scalable. Sprinklr is an early adopter of innovations, and going forward, it will continue to make use of new technologies, such as artificial intelligence–based services.

“Now that we have stable and consistent performance using AWS, we can do things we could never do before,” says Bansal.

About Sprinklr

Founded in 2009, Sprinklr provides businesses around the world with social media, marketing, insight, and customer care services. It has four artificial intelligence–infused product suites in a new category of enterprise software: unified customer experience management.

AWS Services Used

Amazon Keyspaces (for Apache Cassandra)

Amazon Keyspaces (for Apache Cassandra) is a scalable, highly available, and managed Apache Cassandra–compatible database service. With Amazon Keyspaces, you can run your Cassandra workloads on AWS using the same Cassandra application code and developer tools that you use today.

Learn more »

Amazon ElastiCache

Amazon ElastiCache is a fully managed, Redis- and Memcached-compatible service that delivers real-time, cost-optimized performance for modern applications with 99.99% availability.

Learn more »

Amazon Elastic Kubernetes Service

Amazon Elastic Kubernetes Service (Amazon EKS) is a managed Kubernetes service to run e AWS cloud and on-premises data centers.

[Learn more »]

AWS Graviton Processor

AWS Graviton is a family of processors designed to deliver the best price performance for your cloud workloads running in Amazon Elastic Compute Cloud (Amazon EC2).

Learn more »

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